



An Analysis of the First Round of the MLB First-Year Player Draft

Gabriel Chandler & Simon Rosenbaum

To cite this article: Gabriel Chandler & Simon Rosenbaum (2018) An Analysis of the First Round of the MLB First-Year Player Draft, CHANCE, 31:3, 37-43, DOI: [10.1080/09332480.2018.1522211](https://doi.org/10.1080/09332480.2018.1522211)

To link to this article: <https://doi.org/10.1080/09332480.2018.1522211>



Published online: 14 Sep 2018.



Submit your article to this journal [↗](#)



Article views: 230



View related articles [↗](#)



View Crossmark data [↗](#)

An Analysis of the First Round of the MLB First-Year Player Draft

Gabriel Chandler and Simon Rosenbaum

The First-Year Player Draft is a yearly event, held near the beginning of each June. It is the primary mechanism for amateur (primarily high school and college) players to be assigned to the 30 Major League Baseball (MLB) team organizations. It consists of a total of 40 rounds, with draft position being “reverse order”—i.e., teams with worse records from the previous year receive higher placement within the rounds than those with better records.

This is intended to promote competitive balance within the league. Drafted players who opt to turn professional receive a signing bonus, the value of which correlates highly with their draft position, and—nearly always—are assigned to minor league affiliates of the club that drafted them.

After the 2012 season and draft, the MLB implemented a new system governing free agent signings. Once a player becomes eligible for free agency, his team may offer him a “qualifying offer,” which is a single-year contract equal to the average of the top 125 salaries from the previous season.

For the 2017 season, the qualifying offer was approximately \$17 million. Should the player decline the qualifying offer and instead sign with another team, the signing team is required to give up a draft pick. This would be their first-round pick, unless that pick should fall among the first 10 selections. Such picks are known as *protected*.

In the case of a protected pick, the signing team will instead lose their second-round selection. The team from whom the qualifying offer was declined would be compensated with a “sandwich pick” (a selection coming between the first and second rounds of the draft).

It is well understood that higher-round picks are more valuable to a team, since the pool of available talent becomes more shallow as players are selected. For example, MVPs of both the 2015 and 2016 seasons were first-round picks (Josh Donaldson, the 2015 American League MVP, was a supplementary first-round pick; i.e., a sandwich pick). However, the language of the rule privileges the first 10 picks and suggests that there is substantial variability within even the top 30 selections.

The primary goal of this article is to understand how much variability there is within just the first round of the draft, and whether there was justification for protecting the first 10 picks (or whether 10 is just a convenient round number). A simple look at the performance of 10th- versus 11th-pick players, described below, yields strong statistical significance ($p\text{-val} = .003$).

The central technique we use for this analysis is isotonic regression. In the case of a “positive” measure of performance, we search over the class of all decreasing functions, since the explanatory variable is draft slot. Linearity does



Isotonic regression, like linear regression, attempts to find a curve that lies “closest” to a given data set, where closest is defined in terms of sums of squared deviations. However, rather than searching within the class of linear functions, here we only require that the resulting curve is monotone (non-decreasing), so our estimate is non-parametric. Note that a non-increasing curve, as used here, can be found by fitting the negative of the response variable.

not seem an appropriate model for this relationship, since we might not make much distinction between a 299th and 300th pick, but would for a first versus second pick. Thus, the class of decreasing functions seems particularly well suited.

No team should favor a later pick over an earlier one with an expectation to do better, since the set of players available at the later pick are nested inside the set at the earlier pick. As teams attempt to pick the best players possible at each step, it seems reasonable that the expected value of a player should, thus, decrease in the draft slot.

We acknowledge that occasionally a team will pass on a player due to concern about their ability to sign him. We suspect this happens relatively rarely and across many draft slots, and thus would show up in our model primarily as added noise.

Two assumptions are obvious and implicit with using a monotonic function to describe the relationship between draft slot and production. The first is that particular teams are not better than others at predicting productivity and the second is that teams are not drafting for need.

Regarding the former, we examine this at the end of the article and find little evidence of such an

effect. Combined with the fact that draft position for a team changes significantly from year to year, any differences would manifest themselves primarily as noise in our analysis rather than signal. We discuss the issue of drafting for need in the analysis section.

To our knowledge, this is a novel application of order-restricted inference in the analysis of sports data, although it seems natural in many settings here. For instance, player value might be thought of as unimodal in age (monotone on either side of the mode). In addition to an estimate of the expected performance as a function of draft slot, we also look at the significance of differences in the regression function.

Data

We use as our metric of performance Wins Above Replacement (WAR, specifically Sean Smith's rWAR, publicly available at *baseball-reference.com*), which attempts to summarize the total contribution of a player through all aspects of the game, and allows for the comparison of pitchers with position players. A replacement player is defined as someone a team could acquire easily and for minimal cost. WAR then attempts to measure how many additional games a team won due to having this player rather than a replacement.

Draft data were collected from the June draft each year from 1976 to 2000. Although data are available beyond 2000, we stopped here because later drafts have players for whom career WAR is still very much in flux. We consider the first $k = 50$ picks from each of those years. Entries for players who never reached the major leagues are missing. This is true for 491 of the 1,250 entries ($\approx 39\%$), giving an indication of how difficult it is

to predict future performance of amateur baseball players.

Figure 1 shows how this proportion varies from year to year. There is some indication that the addition of the Mariners and Blue Jays in 1977 caused the success rate to increase (as one would expect with the need for an additional 50 major leaguers), but no such effect is apparent for the 1993 or 1998 expansion. This may be due to the increase in international players (non-USA/Canada), who are not eligible for June draft, but have become increasingly common on major league rosters (currently more than 25%).

There is a clear need for data imputation for the researcher to be able to handle the missing data values. While a value of 0 would seem sensible, nearly a third of players (251 of 759) who reached the major leagues had WAR values that were negative. It seems reasonable that a player who reached the major leagues was at least as valuable as a player who did not. Thus, we assigned all draftees who failed to reach the majors a value of -4, the smallest observed record in the data set (obtained by pitcher Brian Williams, the 31st pick in the 1990 draft). We acknowledge that our choice is debatable, with other choices likely to yield somewhat different results.

Analysis

Before fitting a monotonic function to the data, we attempt to allay fears regarding possible non-monotonicity in the relationship, due to teams drafting according to need. Even if teams had the same information available and used it efficiently, there is concern that teams might draft a player who is not the top player available because of positional need (or lack thereof). We examine this by considering how likely a team is

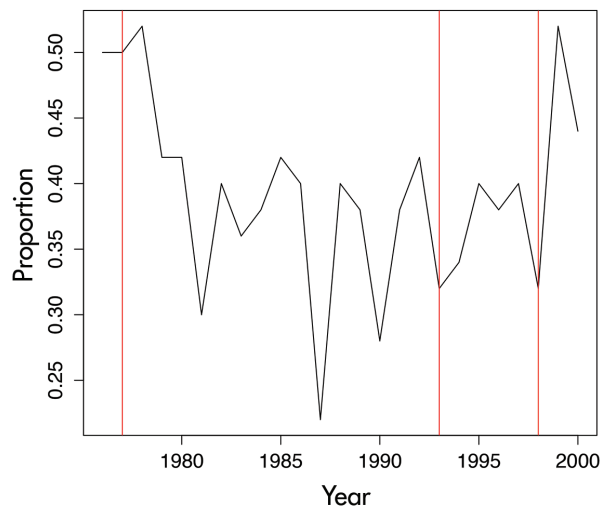


Figure 1. Proportion of drafted players who failed to reach the major leagues, by year. Vertical lines correspond to expansion years (two teams added each occasion).

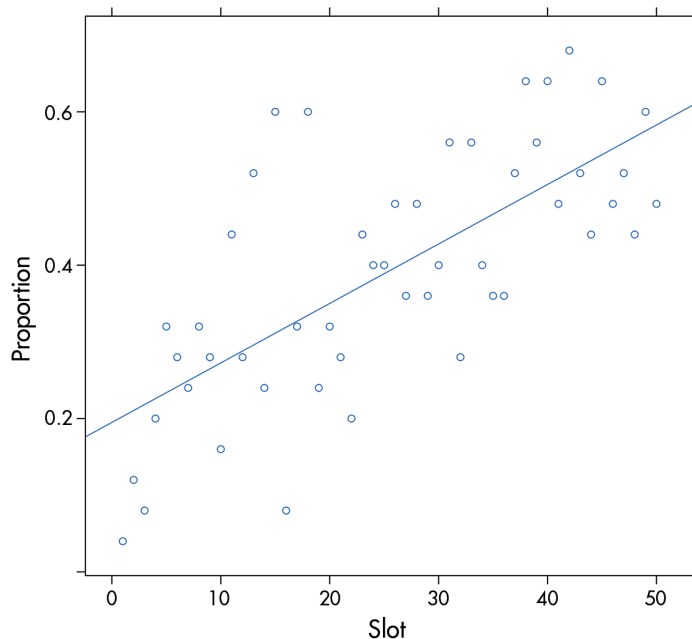


Figure 2. Proportion of drafted players who failed to reach the major leagues, by draft slot, with linear least squares fit.

to draft a pitcher given their team ERA (which indicates how much they need pitching help).

We specifically model the probability a team drafts a pitcher with their first pick (since they may have multiple picks in the first round) against their team ERA. With a sample size of $n = 985$, a logistic regression yields a $\hat{\beta}_1 = 0.157$ ($p < 0.01$) which corresponds to an odds ratio of 117%, meaning

that a team with an ERA a full point higher (a difference of 1.78 standard deviations) only has a 17 percent greater chance of drafting a pitcher. If one looks only in the last decade, $\hat{\beta} = 0.059$, which is highly non-significant. We conjecture this is at least partly due to the extended minor league grooming of almost all players.

We also note that ERA has a very weak correlation with draft

position, again meaning that any effect on our analysis will be seen primarily as added noise and less power of subsequent hypothesis tests.

The natural expectation is that higher-drafted players would have a higher likelihood of a productive major league career than players drafted later. Figure 2 shows the proportion of each draft slot in not reaching the major leagues.

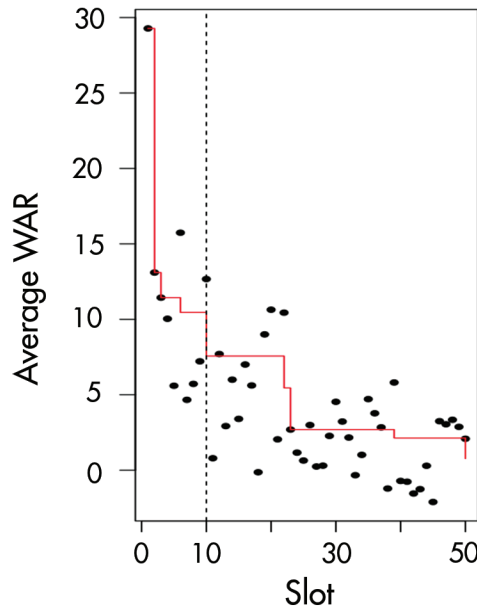


Figure 3. Average (over years) WAR versus draft slot and resulting isotonic regression curve. Dashed vertical line at slot 10, the largest protected pick. Note that the function is continuous from the left.

A linear fit exhibits a lack of fit for very early picks, especially the first three overall selections. One may recognize the names Steve Chilcott and Brien Taylor, the only two first overall picks (1966 and 1991 respectively) not to reach the major leagues.

Of course, drafting a player who simply reaches the major leagues should not be the goal of a team in the first round, so the preceding figure does not capture the whole story. With signing bonuses in the millions of dollars for first-round draft choices, the expectation is for a productive major leaguer. Thus, we seek to understand the relationship between draft slot and expected WAR.

As argued in the introduction, this relationship is clearly monotonically decreasing, due to the set of players available for draft at any slot being a decreasing nested collection of sets. We seek to find the best (in the sense of least squares) monotonically decreasing function

for our data. That is, we consider the WAR at pick i to be of the form $Y_{ij} = \mu_i + \epsilon_{ij}$, with the constraint that $\mu_i \geq \mu_{i+1}$ for all i . In other words, the expected WAR decreases as the draft slot increases.

As in linear regression, we seek the solution that minimizes the sum of squared residuals. Wonderfully, it turns out that this solution can be found rather simply, despite the need to search over a much bigger class of functions than linear regression. The resulting estimated regression function is presented in Figure 3. We see the regression function make a large jump at pick 10, which is the last protected pick under the current MLB rules.

One may conjecture that this feature in the data led to the rule. This jump is largely due to the poor average performance of 11th picks (the worst of any pick among the first 17) and the surprising success of 10th picks (higher than six of the preceding nine slots). One should ask how much of the

observed effect is due to chance. Is there strong statistical evidence to suggest that something interesting is happening between the 10th and 11th picks?

As a simple analysis, we run a two sample t-test with the alternative hypothesis of $H_1: \mu_{10} > \mu_{11}$, which returns a p -value of .003 and a 95% lower confidence bound of 4.95 (note the severe lack of normality in the data, however). Whether an increase in career WAR of 4.95 is meaningful is an open question, as well as whether a more-suitable definition of protected picks may exist. One might also wonder whether this result is a consequence of “data snooping.”

To this end, we wish to construct hypothesis tests for decreasing (versus a null of constancy) value over subsets of picks.

Consider picks c to d . The null hypothesis states that the expected WAR is constant over these picks. In other words, after removing the top $c - 1$ prospects from the draft,

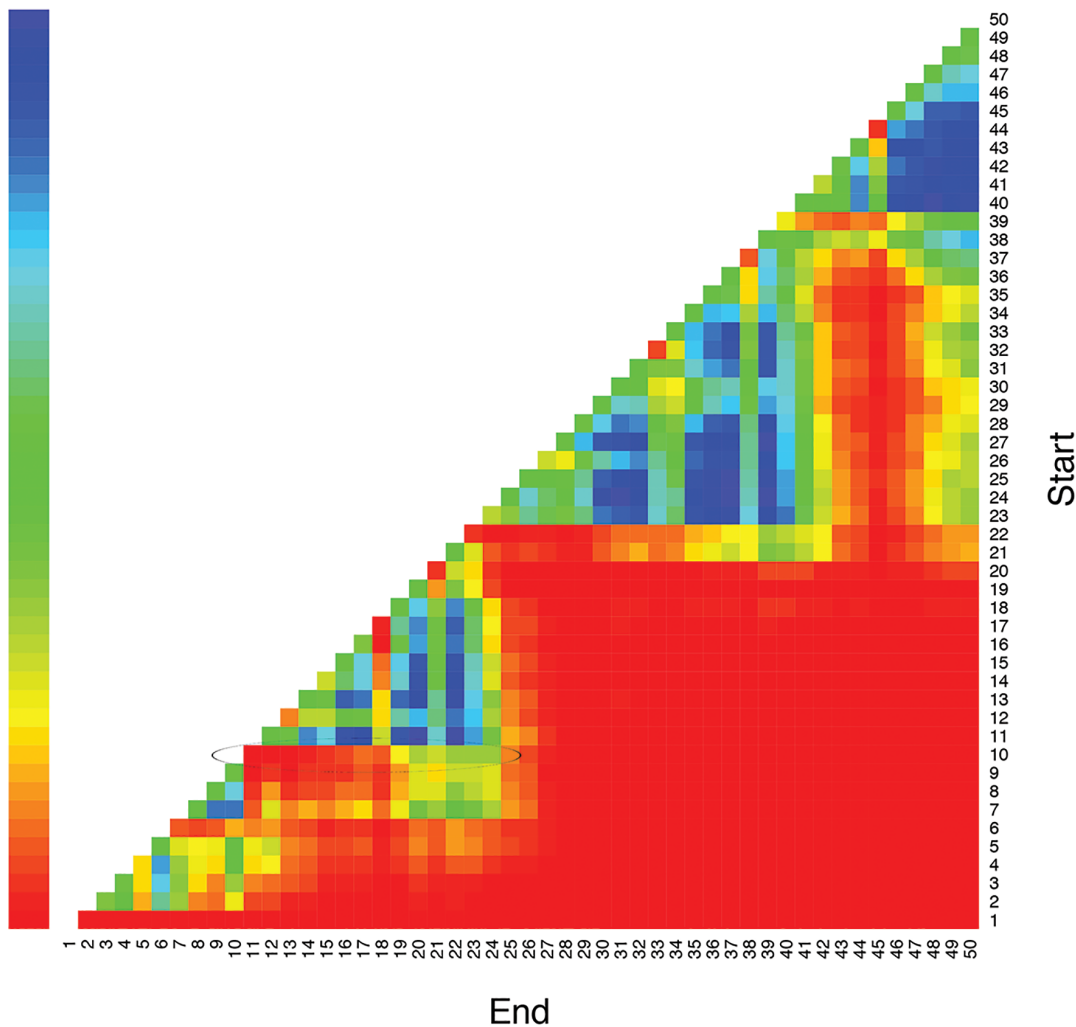


Figure 4. Heat map of p -values for test of constancy of response curve over different ranges of draft slots. Scale goes from 0 (red) to 1 (blue). Tests comparing 10th pick to later picks are circled.

there are at least $d - c + 1$ players who are indistinguishable in terms of their “projectability” and agreed to be the best prospects remaining in the draft class.

In such a scenario, one can view these picks as teams essentially randomly selecting one of these players. We can then simulate from this, by simply taking all players drafted between picks c and d (inclusive) and permuting their draft slot. We then use isotonic regression to estimate the response curve, and see how far it falls from the true response curve

(which under the null is constant). Similarly, we can compute this distance for the unpermuted data, and compare the two. The proportion of time the permuted data results in a distance greater than the original data is an estimate of the p -value.

Figure 4 displays the results of our permutation scheme on the data set, with $m = 1,000$ permutations done for each test. We can see that there is evidence of a difference between the 10th and 11th picks ($p = 0.002$), but this seeming difference is short-lived. There is

not significant evidence to claim that the response curve is decreasing on the set $[10, t]$ for $t = 16, \dots, 26$, even without accounting for multiple testing.

As we might have expected from the discussion thus far, the first pick is special (although interestingly Ken Griffey, Jr. was the first Hall of Fame inductee who was drafted first overall), since the expected WAR is statistically significantly higher than any other pick.

Another interesting observation is the large block of red in the

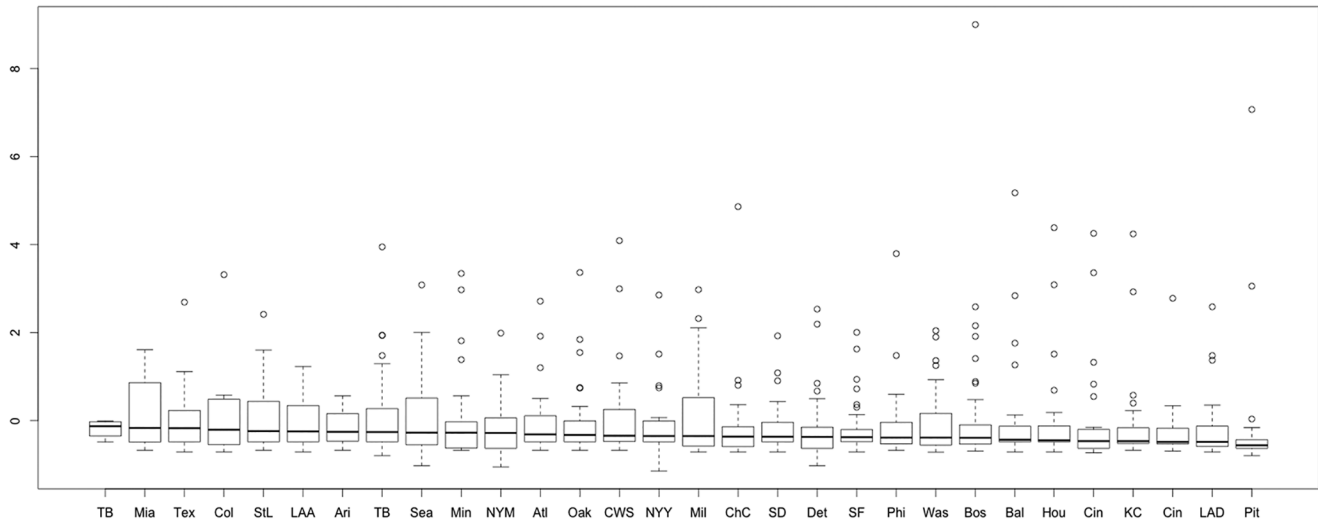


Figure 5. Box plot of z-scores for each player in data set, ordered by 75th percentile. Note: Tampa Bay and Arizona had only four picks each in the data set.

lower right-hand corner of Figure 4. The block covers starting picks from about 1 through 20, and ending picks between 27 and 50. This suggests that any pick in the top 20 should be preferred over any pick 27 or later. Thus, if we were to subscribe to the idea that some picks should be protected regarding free agent signings, a choice somewhere in the range of 20 seems empirically justified.

While the top two thirds of first-round selections may seem like a lot, it would protect the first round pick of exactly the teams to not make the playoffs the previous year (including wild cards, 10 out of 30 teams make the playoffs). This would remove the deterrent of a forfeited first-round draft pick for all non-playoff teams, which would seem to make them more likely to pursue these types of free agents. In debating whether to sacrifice the present for the future, or vice-versa, this sacrifice would be lessened under this definition for such teams.

With these teams improving, the competitive balance of the MLB would increase, which is one of the goals of a reverse-order draft.

Interestingly, the most-recent version of the collective bargaining agreement, announced in late 2016, does away with the protected picks of 1 through 10. Rather, signing teams will now lose either a second- and fifth-round pick, or just a third-round pick, depending on their total team salaries.

Conclusion

We have explored the value of draft position in the case of the MLB June draft. In particular, we were interested in the justification for the definition of “protected” picks being the first 10 picks in the draft. We conjecture that this value came about in one of two ways.

The first is simply acknowledging that earlier picks are more valuable (and this is certainly true for the first pick overall), and then choosing 10 because it is a “nice” number. The second is that

some empirical study was actually conducted, but may have been short-sighted in scope. Historically, 10th picks have done better than their counterparts one pick later, and the result is statistically significant. However, this does not hold once later picks have been incorporated, since there is no significant evidence that the expected WAR is decreasing on picks 10 through 26.

Had statistical significance been present, the obvious follow-up question would have been whether the difference in WAR was also a meaningful difference. While avoiding this question, we instead advocate for protected picks to include the first 20. This both has a better empirical justification and benefits only non-playoff teams, which promotes competitiveness.

Acknowledging that there is a difference within even the first 50 picks of the draft, we might wonder if there is evidence that one team is doing a better job with their high picks than another. Without this observation, we may

conclude that the Seattle Mariners are draft geniuses with Ken Griffey, Jr. (1987, 83.6 career WAR) and Alex Rodriguez (1993, 118.9 career WAR up to end of 2015 season). These selections seem less impressive, however, when we account for the fact that they had the first overall pick for both of these years. To examine this, we computed z-scores for each player in the data set, and grouped them by drafting team.

The standard deviation, assumed to be changing by pick, was also estimated via an isotonic regression (we assume the variability is decreasing in pick number, which was empirically confirmed). Figure 5 displays comparative box plots for each team in terms of their draftees. We note the largest outlier as Roger Clemens, whose 139.4 WAR was very unusual for a 19th pick.

This data set is also in the “pre-Moneyball” era, which suggests that, given modern analytics, the players in our data set may have been drafted in a considerably different order, again altering the results. It will be interesting to compare these results with those of the modern era, once those data become available. Finally, one can reasonably expect somewhat different results should a metric of performance other than career WAR be used. 

Further Reading

Barlow, R.E., Bartholomew, D.J., Bremner, J.M., and Brunk, H.D. 1972. *Statistical inference under order restrictions. The theory and application of isotonic regression*. London-New York-Sydney: Wiley Series in Probability and Mathematical Statistics.

Burger, J.D., and Walters, S.J.K. 2009. Uncertain prospects: rates of return in the baseball draft, *Journal of Sports Economics* 10, 485–501 (draft). All Theses. Paper 2088. http://tigerprints.clemson.edu/all_theses/2088.

Spurr, S.J. 2000. The baseball draft: A study of the ability to find talent, *Journal of Sports Economics* 1, 66–85.

About the Authors

Gabe Chandler is an associate professor in the Department of Mathematics at Pomona College.

Simon Rosenbaum, a 2016 graduate of Pomona College in mathematics and economics and a 2014 NCAA Division-III first team All-American baseball player, is an assistant in baseball development for the Tampa Bay Rays.